

GPRS PROTOCOL

FOR AIS140

SL. #	Version	Prepared by	Date
1	AIS140_GPS_PROTO_V.1 .4	Vijith V.Nair	07/06/2019

CONFIDENTIAL

Introduction

The device will send following type of data to the server/control center.

1. LOGIN Packet
2. Health Packet
3. Location Packet
4. Parameter Change Packet
5. Emergency Alert Packet

The device will send data as ASCII values with comma (,) separation.

PACKETS FROM VTL TO SERVER

LOGIN Packet

The VLT will send a login message as per Table1 format whenever it makes new connection with the server and no response or acknowledgement is expected from the server.

Table-1		
Field	Description	Sample
\$	Start character	\$
VRN	Vehicle registration number as device name	GA01AA1234
IMEI	15 digit unique ID of device	862634571200321
Firmware Version	Device firmware version	V1.1.1
Protocol Version	Protocol version of device	VERSION1.0
Lat	Last valid Latitude	28.758963
Latitude Direction	Last valid Latitude Direction	N
Long	Last valid Longitude	77.6277844
Long. Direction	Last valid longitude direction	E

Example

\$,GA01AA1234,864495039240301,V1.1.1,VERSION1.0,28.359791,N,76.927086,E

Health Packet

The VLT would send a health packet to server at set intervals as per format in Table-2 and this threshold value is configurable over the air.

Table-2			
SL.No.	Field	Description	Sample
1	Start Character	\$	\$
2	Header	The header of The packet/identifier	HP
3	Vendor ID	Vendor identification header	ASPL
4	Firmware Version	Version details of the Firmware Used	V1.1.1
5	IMEI	Identified of the sending unit.15 Digit standard unique IMEI no.	862634571200321
6	Battery percentage	Indicates the internal battery Level	100
7	Low battery threshold Value	Indicates value on which low Battery alert generated in percentage	15
8	Memory percentage	Indicates flash memory percentage used	12
9	Data update rate when IGNITION is ON	Indicates Packet frequency on IGNITION ON in seconds	10
10	Data update rate when IGNITION is OFF	Indicates Packet frequency on IGNITION OFF in seconds	60
11	Digital I/o status	Status of Inputs connected to the device(I1I2I3I4) Refer Table-3 for details	1111
12	Analog I/o status-A1	Analog input-1 status in Volts(Main battery Voltage)	11.6
13	Analog I/o status-A2	Analog input-2 status in Volts	7.5
14	End character	*	*

INPUT Status

Table-3		
INPUTs	Description	Values
I1	High Sense	0=ON/connected, 1= OFF/Disconnected
I2	Low Sense	1=ON/connected, 0= OFF/Disconnected
I3	Low Sense	0=ON/connected, 1= OFF/Disconnected
I4	Low Sense	0=ON/connected, 1= OFF/Disconnected

Example

\$HP,ASPL, 1.4D3_AIS140_1.0,862634571200321,100,15,12,10,60,1111,11.6,7.5,*

Location Message Format

This is the format of location and alert data sending to server

Table-4			
SL.No	Field	Description	Sample
1	Start Character	\$	\$
2	Header	The header of the packet/identifier	CP
3	Vendor ID	Vendor identification header	ASPL
4	Firmware Version	Firmware version of device	V1.1.1
5	Packet type	See Table-5 for details	NR
6	Packet ID	See Table-5 for details	01
7	Packet Status	L=Live or H=History	L
8	IMEI	Identifier of the sending unit.15 digit standard	8626123659870012
9	VRN	Vehicle registration number	GA01AA1234
10	GPS Fix	1= available, 0= Void (GPS signal availability)	1
11	Date	Date as per GPS in ddmmyyyy format	22072018
12	Time	Time as per GPS in UTC format(hhmmss)	052517
13	Latitude	Latitude in decimal degrees	28.758963
14	Latitude Dir.	Latitude Direction N/S(North/South)	N
15	Longitude	Longitude in decimal degrees	77.6277844
16	Longitude Dir.	Longitude Direction E/W (East/West)	W

17	Speed	Speed in kmph up to one decimal value	25.5
18	Heading	True course over ground in degrees	275.8
19	No of Satellites	Number of satellites Available for fix	8
20	Altitude	Altitude of the device in Meters	250.5
21	PDOP	Positional dilution of Precision	1.2
22	HDOP	Horizontal dilution of Precision	1.0
23	Operator name	GSM network Operator Name	Airtel
24	Ignition Status	1=ON, 0=OFF	1
25	Main Power status	1=ON, 0=OFF	1
26	Main Battery Voltage	Indicator showing source Voltage in Volts.(Upto One Decimal Value)	12.3
27	Internal Battery Voltage	Internal battery voltage in Volts.	4.2
28	Emergency status(Panic)	1=ON, 0=OFF	0
29	Temper Alert	C= cover closed, O =Cover open	C
30	GSM signal strength	Value from 0-31	28
31	MCC	Mobile country code	404
32	MNC	Mobile network code	13
33	LAC	Local area code	00D6
34	Cell ID	GSM Cell ID	CFBD
35	NMR	Neighboring 4 cellID Along with their LAC & signal strength	
36	Digital input status	Status of Inputs connected to the device(I1I2I3I4) Refer Table-3	0111

		for details	
37	Digital output	Digital output status O1O2 (0=OFF, 1=ON)	00
38	Frame Number	Message sequence 000001 to 999999	000012
39	Odometer Reading	Distance covered in Km	124.5
40	Current Fence	Fence ID in which the Vehicle is present	A
41	RESERVED FIELD	RESERVED FOR FUTURE	
42	Checksum	Checksum of packet	!
43	End character	(END of packet) *	*

Protocol Structure

START CHARATER, *HEADER*, Vendor ID, Firmware Version, Packet type, Packet ID, Packet status, IMEI, VRN, GPS Fix, Date, Time, Latitude, Latitude Dir, Longitude, Longitude Dir, Speed, Heading, No. Of satellite, Altitude, PDOP, HDOP, Network operator, IGNITION, Main Power, Main battery Voltage, Internal battery Voltage, Emergency, Temper alert, GSM signal strength, MCC, MNC, LAC, CellID, CellID1, LAC1, SS1, CellID2, LAC2, SS2, CellID3, LAC3, SS3, CellID4, LAC4, SS4, I1I2I3I4, O1O2, Frame number, Odometer, FenceID, CHECKSUM*,

EXAMPLE

1. Live Data

\$,CP,ASPL,V1.1.1,NR,01,L,864495039252728,DL1RQ2455,1,21082018,102209,28.359692,N,76.927437,E,0.0,36.00,8,269.0,1.2,1.7,Airtel P
N,0,1,12.2,4.0,0,C,25,404,10,209,c342,6d17,209,24,d2fb,209,20,d2fc,209,19,c344,209,19,0111,00,002165,125.4,A,,?*

2. Memory Data

\$,CP,ASPL,V1.1.1,NR,02,H,864495039252728,DL1RQ2455,1,21082018,102209,28.359692,N,76.927437,E,0.0,36.00,8,269.0,1.2,1.7,Airtel P
N,0,1,12.2,4.0,0,C,25,404,10,209,c342,6d17,209,24,d2fb,209,20,d2fc,209,19,c344,209,19,0111,00,002165,125.4,A,,?*

3. Event Packet for IGN ON

\$,CP,ASPL,V1.1.1,IN,07,L,864495039252728,DL1RQ2455,1,21082018,102209,28.359692,N,76.927437,E,0.0,36.00,8,269.0,1.2,1.7,Airtel P
N,1,1,12.2,4.0,0,C,25,404,10,209,c342,6d17,209,24,d2fb,209,20,d2fc,209,19,c344,209,19,0111,00,002165,125.4,A,,?*

Packet Type

Table-5			
Sl#	Type	ID	Description
1	NR	01	Normal Data Live
2	NR	02	Normal Data History
3	BD	03	Battery Disconnected
4	BL	04	Battery Low
5	BH	05	Low Battery Removed (Charged up to 15%)
6	BR	06	Battery Reconnected
7	IN	07	IGNITION ON
8	IF	08	IGNITION OFF
9	TA	09	Tamper Alert/ GPS cover Opened
10	EA	10	Emergency Alert
11	EU	11	Emergency Free Acknowledgement
12	HB	13	Harsh Braking
13	HA	14	Harsh Acceleration
14	RT	15	Rash Turn
15	DT	16	Panic Wire Cut
16	OD	00	Over Speed
17	FIX	21	Geo-Fence IN, X is fence ID
18	FOX	22	Geo-Fence OUT, X is fence ID

Parameter Change Packet Message Format

This is the format for acknowledging the server about any change in the device parameter over the air. The commanding may either as SMS command or as server command. The packet contains the IP or mobile number from which the device received the command and the parameter changed.

Various fields in the packet are

Table-6			
SL.No	Field	Description	Sample
1	Start Character	\$	\$
2	Header	The header of the packet/identifier	CP
3	Vendor ID	Vendor identification header	ASPL
4	Firmware Version	Firmware version of device	V1.1.1
5	Packet type	Parameter change type	PC
6	Packet ID	Parameter ID	12
7	Packet Status	L=Live or H=History	L
8	IMEI	Identifier of the sending unit.15 digit standard	8626123659870012
9	VRN	Vehicle registration number	GA01AA1234
10	GPS Fix	1= available, 0= Void (GPS signal availability)	1
11	Date	Date as per GPS in ddmmyyyy format	22072018
12	Time	Time as per GPS in UTC format(hhmmss)	052517
13	Latitude	Latitude in decimal degrees	28.758963
14	Latitude Dir.	Latitude Direction	N

		N/S(North/South)	
15	Longitude	Longitude in decimal degrees	77.6277844
16	Longitude Dir.	Longitude Direction E/W (East/West)	W
17	Speed	Speed in kmph up to one decimal value	25.5
18	Heading	True course over ground in degrees	275.8
19	No of Satellites	Number of satellites Available for fix	8
20	Altitude	Altitude of the device in Meters	250.5
21	PDOP	Positional dilution of Precision	1.2
22	HDOP	Horizontal dilution of Precision	1.0
23	Operator name	GSM network Operator Name	Airtel
24	Ignition Status	1=ON, 0=OFF	1
25	Main Power status	1=ON, 0=OFF	1
26	Main Battery Voltage	Indicator showing source Voltage in Volts.(Upto One Decimal Value)	12.3
27	Internal Battery Voltage	Internal battery voltage in Volts.	4.2
28	Emergency status(Panic)	1=ON, 0=OFF	0
29	Temper Alert	C= cover closed, O =Cover open	C
30	GSM signal strength	Value from 0-31	28
31	MCC	Mobile country code	404
32	MNC	Mobile network code	13
33	LAC	Local area code	00D6

34	Cell ID	GSM Cell ID	CFBD
35	NMR	Neighboring 4 cellID Along with their LAC & signal strength	
36	Digital input status	Status of Inputs connected to the device(I1I2I3I4) Refer Table-3 for details	0111
37	Digital output	Digital output status O1O2 (0=OFF, 1=ON)	00
38	Frame Number	Message sequence 000001 to 999999	000012
39	Odometer Reading	Distance covered in Km	124.5
40	Current Fence	Fence ID in which the Vehicle is present	A
41	RESERVED FIELD	RESERVED FOR FUTURE	
42	Checksum	Checksum of packet	!
40	End character	(END of packet) *	*
41	Parameter ID	Id of the parameter changed. Refer Table-7 for details	G
42	Action	Command Action G= Get Parameter S= Set Parameter C= Clear Parameter	C
43	Command Source	Ip or mobile number from which the device received the Command	10.11.12.13

Protocol Structure

START CHARATER, *HEADER*, Vendor ID, Firmware Version, Packet type, Packet ID, Packet status, IMEI, VRN, GPS Fix, Date, Time, Latitude, Latitude Dir, Longitude, Longitude Dir, Speed, Heading, No. Of satellite, Altitude, PDOP, HDOP, Network operator, IGNITION, Main Power, Main battery Voltage, Internal battery Voltage, Emergency, Temper alert, GSM signal strength, MCC, MNC, LAC, CellID, CellID1, LAC1, SS1, CellID2, LAC2, SS2, CellID3, LAC3, SS3, CellID4, LAC4, SS4, I1I2I3I4, O1O2, Frame number, Odometer, FenceID
 ,,CHECKSUM*, ParameterID, Action, Command Source

EXAMPLE

1. Live Data

\$,CP,ASPL,V1.1.1,PC,12,L,864495039252728,GA01AA1234,1,24082018,092309,28
 .359901,N,76.927277,E,0.0,21.00,0,0.0,1.5,0.0,Airtel P
 R,1,1,12.2,3.9,0,C,26,404,10,209,6d17,d2fb,209,16,d2fc,209,15,449,209,13,4634,209,11,0111,00
 ,000234,125.4,A,,?*,O,G,+919873569848

2. Memory Data

\$,CP,ASPL,V1.1.1,PC,12,H,864495039252728,GA01AA1234,1,24082018,092309,2
 8.359901,N,76.927277,E,0.0,21.00,0,0.0,1.5,0.0,Airtel P
 R,1,1,12.2,3.9,0,C,26,404,10,209,6d17,d2fb,209,16,d2fc,209,15,449,209,13,4634,209,11,0111,00
 ,000234,125.4,A,,?*,O,G,+919873569848

Parameter ID

Table-7			
SL #	Command Name	Command	LOOKUP CODE
1	Panic clear	PANICRLS	A
2	SMS Tracking	#SMSSERVER::--; #SMSSERVER? #SMSSERVER::number;	B
3	Vehicle ID	VID::veh_id; VID? VID--;	C
4	Harsh Event	HT::HA::HB::angle; HT? HT--;	D
5	Tracking Interval	WEBSTARTign_on_interval,ignoff_interval,main_pwr_off_interval,panic_interval	E
6	PACKET TIME	GMS::time_in_min;	F
7	HEALTH	HPI::TIME;	G

	PACKET INTERVAL	HPI? HPI--	
8	FACTORY RESET	FR	H
9	PASSWORD	PW::PASWORD;	I
10	GPRS STATUS	GPRSSTATUS	J
11	VERSION	VERSION	K
12	GPS CHECK	GETGPS	L
13	APN CONFIGURATI ON	#CONFIG::APN::UN::PW; #CONFIG?	M
14	SYSTEM STATUS	SYSSTATUS	N
15	SERVERCHAN GE	#SERVERCHANGE? #SERVERCHANGE::IP::PORT; #SERVERCHANGE1? #SERVERCHANGE1::IP::PORT;	O
16	OVERSPEED	SPEED? SPEED::--; SPEED::OVERSPEED_LIMIT_I N_KMPH;	P
17	EMERGENCY TIMEOUT	EST? EST-- EST::TIMEOUT_IN_MIN;	Q
18	SLEEP TIME	SLT? SLT-- SLT::SLEEP_INTERVAL_IN_M IN;	R
19	TOGGLE SIM	TSIM	S

20	IMEI	IMEI	T
21	SLEEP ENABLE	SLEEPON	U
22	SLEEP DISABLE	SLEEPOFF	V
23	FLASH ERASE	FE	W

CHECKSUM CALCULATION -1

This method of checksum calculation is used for location packet and parameter change packets.

The checksum calculation method is

```

Char chksum_gprs(char * ptr)
{
    unsignedint i = 0;
    charchksum=0;

    for(i=0;i<1000 && *ptr != 0x00;i++)
    {
        chksum ^= *ptr++;
    }
    return(chksum);
}

```

NOTE:

CHECKSUM OF ALL BYTES IN PACKET BETWEEN STARTING CHARACTER (\$) AND FRAME NUMBER, INCLUDING BOTH.

EMERGENCY ALERT PACKET

On pressing panic button, an Emergency Alert would be generated as per below format. The Emergency packet would be send to two IPs or as SMS to any control sender number if GPRS is not working. The Emergency state can be cleared only from clear command from server or it will clear automatically after the preset interval.

Table-8		
Field	Description	Sample
Start Character	Start Character of the packet	\$
Packet Header	EPB, The unique identifier for all messages from VLT	EPB
Message Type	Message Types supported. Emergency Message(EMR) or Stop Message(SEM)	EMR
Vehicle ID	Unique ID of the Vehicle (IMEI Number)	8626123659870012
Packet Type	NM–Normal Packet, SP–Stored Packet	NM
Date	Date and time of location the location obtained from the data in DDMMYYYYYhhmmss format	12052017053545
GPS Validity	A–Valid, V – Invalid	A
Latitude	Latitude in decimal degrees -dd.mmmmmm format	28.758963
Latitude Direction	N –North, S–South	N
Longitude	Long. in decimal degrees-dd.mmmmmm format	77.6277844
Long. Dir.	E –East, W–West	W
Altitude	Altitude in meters(above sea level in meter)	235.0
Speed	Speed of Vehicle as Calculated by GPS module in VLT.(in km/hr)	25.5
Distance	Distance calculated from previous GPS data(in KM)	0001.50
Provider	G- Fine GPS N-Coarse GPS or data from the network	G
Vehicle RegnNo	Registration Number of Vehicle	GA01AA1234
Reply number	Mobile number in to which Text message to be send(Emergency number provided by MHA/state)	0

*	End character	*
Checksum	Checksum of packet(Optional)	

Example

Emergency Pressed

\$,EPB,EMR,864495039252728,NM,21082018104217,A,28.359692,N,76.927437,E,269,0.0,0.00
,G,GA01AA1234,0,*? ??????

Emergency Pressed-Stored Packet

\$,EPB,EMR,864495039252728,SP,21082018104217,A,28.359692,N,76.927437,E,269,0.0,0.00,
G,GA01AA1234,0,*? ??????

Emergency Alert Removed

\$,EPB,SEM,864495039252728,NM,21082018104217,A,28.359692,N,76.927437,E,269,0.0,0.00
,G,GA01AA1234,0,*? ??????

CHECKSUM CALCULATION -2

This method of checksum calculation is used for EMERGENCY packets.

The checksum calculation method is

```
unsignedlongint chksum_gprs(char * ptr)
{
    unsignedint i = 0;
    unsignedlongint chksum=0;
    for(i=0;i<1000 && *ptr != 0x00;i++)
    {
        chksum ^= *ptr++;
    }
    return(chksum);
}
```

```
char *l_ptr=0;
l_ptr=(char*)&chksum;
if((*l_ptr==0) || (*l_ptr == 0x1B) || (*l_ptr ==0x1A))*l_ptr=0xFF;
g_gen_sharing_buffer[p+i] = *l_ptr++;
if((*l_ptr==0) || (*l_ptr == 0x1B) || (*l_ptr ==0x1A))*l_ptr=0xFF;
g_gen_sharing_buffer[p+i+1] = *l_ptr++;
if((*l_ptr==0) || (*l_ptr == 0x1B) || (*l_ptr ==0x1A))*l_ptr=0xFF;
g_gen_sharing_buffer[p+i+2] = *l_ptr++;
if((*l_ptr==0) || (*l_ptr == 0x1B) || (*l_ptr ==0x1A))*l_ptr=0xFF;
g_gen_sharing_buffer[p+i+3] = *l_ptr++;
if((*l_ptr==0) || (*l_ptr == 0x1B) || (*l_ptr ==0x1A))*l_ptr=0xFF;
g_gen_sharing_buffer[p+i+4] = *l_ptr++;
if((*l_ptr==0) || (*l_ptr == 0x1B) || (*l_ptr ==0x1A))*l_ptr=0xFF;
g_gen_sharing_buffer[p+i+5] = *l_ptr++;
if((*l_ptr==0) || (*l_ptr == 0x1B) || (*l_ptr ==0x1A))*l_ptr=0xFF;
g_gen_sharing_buffer[p+i+6] = *l_ptr++;
if((*l_ptr==0) || (*l_ptr == 0x1B) || (*l_ptr ==0x1A))*l_ptr=0xFF;
g_gen_sharing_buffer[p+i+7] = *l_ptr++;
```

NOTE:

CHECKSUM OF ALL BYTES IN PACKET BETWEEN STARTING CHARACTER (\$) AND END CHARACTRT (*) INCLUDING BOTH.

RESPONSE TO ACTIVATION AND HEALTH CHECK COMMAND

The device will send reply to activation and health check commands in the following format as a TEST message as per below Table -9. The separator used is comma (,)

Table-9		
Field	HEALTH CHECK REPLY	ACTIVATION REPLY
Header	HCHKR	ACTVR
Random code	343434	474747
Vendor ID	ASPL	ASPL
Firmware Version	V1.1.1	V1.1.1
IMEI	8626123659870012	8626123659870012
Alert ID	1	1
Latitude	28.758963	28.758963
Latitude Dir.	N	N
Longitude	77.6277844	77.6277844
Longitude Dir.	W	W
GPS Fix	1	1
Date &Time	05042019 120317	05042019 120317
Heading	275.80	275.80
Speed*	25.4	25.4
GSM signal strength	28	28
MCC	404	404
MNC	10	10
LAC	d6d6	d6d6
Main Power	1	1
IGN STATUS	1	1
Battery Voltage	11.8	11.8
Frame number	100000	100000
Vehicle Mode	ID	ID

Example**Activation Response**

ACTVR,343434,ASPL,1.4D3_AIS140_1.1,864495039252728,1, 28.359692,N,76.927437,E,1, 31032019
052636,269,0.0,28,404,04,03B6,1,0,19.2,000056,ID

Health Check Response

HCHKR,343434,ASPL,1.4D3_AIS140_1.1,864495039252728,1, 28.359692,N,76.927437,E,1, 31032019
053632,269,0.0,28,404,04,03B6,1,0,19.2,000057,ID

GPRS command

GPRS commanding can be used to set the parameters of the device. The command format would be as follows

Table-10		
Field	Description	Sample
Header	Header of the packet	\$MSG
Parameter	Parameter with proper format and password	IMEI<6906>
End character	End character of device	&

Example of Structure

\$MSG, command<password>&

Example

\$MSG,IMEI<6906>&

This command will give you the IMEI number, 6906 is the default device password.

Note: - Refer Instruction manual for all command list.

GPRS COMMAND RESPONSE

The response from the device for the command would be in following format

Table-11		
Field	Description	Sample
Header	Header of the packet	\$
IMEI	IMEI number of device	864495039252728
Response	Response for command	864495039252728
End character	End character of device	&&

Example

The response for IMEI command would be

\$864495039252728, 864495039252728&&

Note: - Refer Instruction manual for all command list.

Note:-

1. Location based on network in Emergency packet is being fetched from Google API
2. Unit of Speed in all packets are ***kmph*** and the time is in GMT